

Application to the fundamental group

SGA 2, Exposé X

English draft

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About this draft

This is an unofficial English translation of Exposé X, “Application to the fundamental group”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ X

Application to the fundamental group

Throughout this exposé, one will denote by X a locally noetherian prescheme, by Y a closed subset of X , by U a variable open neighbourhood of Y in X and by $\widehat{X}$ the formal completion of X along Y (EGA I 10.8). For every prescheme Z , one will denote by $\mathbf{Et}(Z)$ the category of étale coverings of Z , and by $\mathbf{L}(Z)$ the category of locally free coherent Modules on Z .

1. Comparison of $\mathbf{Et}(\widehat{X})$ and of $\mathbf{Et}(Y)$

Let $\mathcal{I}$ be an ideal of definition of Y in X . Set, for every $n \in \mathbf{N}$, $Y_n = (Y, (\mathcal{O}_X/\mathcal{I}^{n+1})|_Y)$. The Y_n form an inductive system of ordinary preschemes, or also of formal preschemes, by endowing the structure sheaves with the discrete topology. One knows (EGA I 10.6.2) that $\widehat{X}$ is the inductive limit, in the category of formal preschemes, of the inductive system of the Y_n . One also knows (EGA I 10.13) that to give a formal $\widehat{X}$ -prescheme of *finite type* R is to give an inductive system of ordinary Y_n -preschemes R_n of *finite type*, such that $R_n \simeq (R_{n+1}) \times_{(Y_{n+1})} (Y_n)$. Moreover, for R to be an étale covering of $\widehat{X}$, it is necessary and sufficient that for every n , R_n be an étale covering of Y_n . This being said, it is easy to see that nilpotent elements do not matter for étale coverings (SGA 1 8.3), i.e. that the change of base functor:

$$\mathbf{Et}(Y_{n+1}) \longrightarrow \mathbf{Et}(Y_n)$$

is an equivalence of categories for every $n \in \mathbf{N}$. Hence:

PROPOSITION 1.1. *With the notations introduced above, the natural functor $\mathbf{Et}(\widehat{X}) \rightarrow \mathbf{Et}(Y)$ is an equivalence of categories (cf. SGA 1 8.4).*

2. Comparison of $\mathbf{Et}(Y)$ and $\mathbf{Et}(U)$, for U variable

We shall introduce two conditions from which the announced comparison theorem will follow easily. Let X be a locally noetherian prescheme and let Y be a closed subset of X . One says that the pair (X, Y) satisfies the *Lefschetz condition*, which one writes $\text{Lef}(X, Y)$, if, for every open U of X containing Y and every locally free coherent sheaf E on U , the natural homomorphism

$$\Gamma(U, E) \longrightarrow \Gamma(\widehat{X}, \widehat{E})$$

is an isomorphism.

One says that the pair (X, Y) satisfies the *effective Lefschetz condition*, which one writes $\text{Leff}(X, Y)$, if one has $\text{Lef}(X, Y)$ and if moreover, for every locally free coherent sheaf $\mathcal{E}$ on $\widehat{X}$, there exist an open neighbourhood U of Y and a locally free coherent sheaf E on U and an isomorphism $\widehat{E} \simeq \mathcal{E}$.

These conditions are satisfied in two important examples:

EXAMPLE 2.1¹. Let A be a noetherian ring and let $t \in \mathfrak{r}(A)$ be an A -regular element belonging to the radical $\mathfrak{r}(A)$ of A . Suppose that A is a quotient of a regular local ring and that A is complete for the t -adic topology (for example A complete for the $\mathfrak{r}(A)$ -adic topology). Set $X' = \mathrm{Spec}(A)$ and $Y' = V(t)$, and moreover set $x = \mathfrak{r}(A)$ and $X = X' - \{x\}$, $Y = Y' - \{x\}$. Thus X is open in X' and $Y = X \cap Y'$. Then:

- (i) If, for every prime ideal $\mathfrak{p}$ of A such that $\dim A/\mathfrak{p} = 1$ (i.e. for every closed point of X) one has $\mathrm{prof} A_{\mathfrak{p}} \geq 2$, then one has $\mathrm{Lef}(X, Y)$;
- (ii) if moreover, for every prime ideal $\mathfrak{p}$ of A such that $t \in \mathfrak{p}$ and $\dim A/\mathfrak{p} = 1$ (i.e. for every closed point of Y), one has $\mathrm{prof} A_{\mathfrak{p}} \geq 3$, then one has $\mathrm{Leff}(X, Y)$.

Let us first show that for every open neighbourhood U of Y in X , the complement of U in X is a union of finitely many closed points (in X). Let us remark that U is open in X , hence in X' , hence $Z' = X' - U$ is closed. Let I be an ideal of definition of Z' ; it suffices to prove that A/I is of dimension 1. Now $Z' \cap Y' = \{x\}$ hence $A/(I + (t))$ is artinian, whence the conclusion thanks to the ‘‘Hauptidealsatz’’.

The first hypothesis is equivalent to: ‘‘for every prime ideal $\mathfrak{p}$ of A , $\mathfrak{p} \neq \mathfrak{r}(A)$, one has $\mathrm{prof} A_{\mathfrak{p}} \geq 3 - \dim A/\mathfrak{p}$ ’’. Indeed A is a quotient of a regular ring, one can therefore apply ?? ?? to the prescheme X' , to the closed subset $\{x\}$ and to the coherent sheaf $\mathcal{O}_{X'}$, observing that, $c(\mathfrak{p}) = \dim(A/\mathfrak{p})$ for $\mathfrak{p} \in U = X' - \{x\}$ (because x is the closed point of X').

Let U be an open neighbourhood of Y in X and E a locally free $\mathcal{O}_U$ -module. Set $Z = X - U$ and let $u: U \rightarrow X$ be the canonical immersion. One will first prove that $u_*(E)$ is a coherent $\mathcal{O}_X$ -Module, or, what amounts to the same, that $\underline{H}_Z^i(E')$ is coherent for $i = 0, 1$, where E' is a coherent extension of E to X . For this one applies theorem ?? ?? to the prescheme X , to the closed subset Z and to the coherent sheaf E' . It suffices to verify that for every point $p \in U$ such that $c(p) = 1$, one has $\mathrm{prof} E'_p \geq 1$, where one has set

$$c(p) = \mathrm{codim}(\overline{\{p\}} \cap Z, \overline{\{p\}}).$$

Now if $p \in U$ and if $c(p) = 1$, denoting by $\mathfrak{p}$ the ideal of A corresponding to p , one sees that $\dim A/\mathfrak{p} = 2$, because the complement of U is a union of finitely many closed points and A is a quotient of a regular ring. Moreover, E is locally free, hence, for every $p \in \mathrm{Supp} E$, one has $\mathrm{prof} E_p = \mathrm{prof} \mathcal{O}_{U,p}$. Finally, if $p \in U$ and if $c(p) = 1$, one has

$$\mathrm{prof} E'_p = \mathrm{prof} E_p = \mathrm{prof} \mathcal{O}_{U,p} = \mathrm{prof} A_{\mathfrak{p}} \geq 3 - 2 = 1.$$

We must now prove that the natural homomorphism

$$(1) \quad \Gamma(U, E) \longrightarrow \Gamma(\widehat{X}, \widehat{E})$$

is an isomorphism. Setting then $\overline{E} = u_*(E)$, one notes that $\overline{E}$ is coherent and of depth ≥ 2 at all the closed points of X . It follows that $R^i f_*(\overline{E})$ is coherent for $i = 0, 1$, where $f: X \rightarrow X'$ denotes the canonical immersion of X in $X' = \mathrm{Spec}(A)$ (Exp. ??). One then applies (?? ??), and one concludes that

$$(2) \quad \Gamma(U, \overline{E}) \longrightarrow \Gamma(\widehat{X}, \widehat{\overline{E}})$$

is an isomorphism, because A is complete for the t -adic topology.

One has a commutative diagram:

$$\begin{array}{ccc} \Gamma(X, \overline{E}) & \xrightarrow{\simeq} & \Gamma(U, E) \\ & \searrow \simeq & \swarrow \\ & \Gamma(\widehat{X}, \widehat{E}) & \end{array}$$

¹**N.D.E.** one can slightly improve (i): see Mme Raynaud (Raynaud M., ‘‘Théorèmes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental’’, *Ann. Sci. Éc. Norm. Sup. (4)* **7** (1974), p. 29–52, corollaries I.1.4 and I.5); condition (ii) can be improved so as to get rid of the depth conditions along Y (see theorem 3.3 of *loc. cit.* for a precise statement). The proof of this last point is very technical, the article above moreover giving only indications of the proof, referring to a detailed earlier version, which appeared in the Bulletin de la Société mathématique de France.

whence the conclusion.

Let now $\mathcal{E}$ be a locally free coherent sheaf on $\widehat{X}$. If one has proved that $\mathcal{E}$ is algebraizable, i.e. is isomorphic to the formal completion of a coherent $\mathcal{O}_X$ -Module E , it is easy to see that E is locally free in a neighbourhood of Y , hence to prove $\text{Leff}(X, Y)$. Let $\widehat{X}'$ be the formal spectrum of A for the t -adic topology, which is identified with the formal completion of X' along Y' . Let us denote by f the canonical immersion of X in X' , by f' the canonical immersion of Y in Y' , and by $\widehat{f}$ the morphism deduced by passage to the completions. For $\mathcal{E}$ to be algebraizable it suffices that $\widehat{f}_*(\mathcal{E})$ be a coherent $\mathcal{O}_{\widehat{X}}$ -Module, because A is complete for the t -adic topology. Let $\mathcal{I} = t\mathcal{O}_{\widehat{X}}$, it is an ideal of definition of $\widehat{X}$.

For every $n \geq 0$, set $E_n = \mathcal{E}/\mathcal{I}^{n+1}\mathcal{E}$. At every closed point $y \in Y$, the depth of E_0 is ≥ 2 ; indeed t is an A -regular element hence $\text{prof } \mathcal{O}_{Y_0, y} = \text{prof } \mathcal{O}_{X, y} - 1 \geq 2$. One concludes that $\widehat{f}_*(\mathcal{E})$ is coherent (?? ??). qed

EXAMPLE 2.2. (Will allow one to compare the fundamental group of a projective variety and of a hyperplane section) *Let K be a field and let X be a proper K -prescheme. Let $\mathcal{L}$ be an ample invertible $\mathcal{O}_X$ -Module. Let $t \in \Gamma(X, \mathcal{L})$ be an $\mathcal{O}_X$ -regular element, which means that, for every open U and every isomorphism $u: \mathcal{L}|_U \rightarrow \mathcal{O}_U$, $u(t)$ is not a zero-divisor in $\mathcal{O}_U$ (a condition which does not depend on u). Let $Y = V(t)$ be the subscheme of X with equation $t = 0$.² Then:*

- (i) *If, for every closed point x in X , one has $\text{prof } \mathcal{O}_{X, x} \geq 2$, one has $\text{Lef}(X, Y)$;*
- (ii) *if moreover, for every closed point $y \in Y$, one has $\text{prof } \mathcal{O}_{X, y} \geq 3$, one has $\text{Leff}(X, Y)$.*

This example will be treated in detail in Exp. ??.

Let S be a prescheme, one knows (EGA II 6.1.2) that the functor which, to every finite flat covering $r: R \rightarrow S$, associates the $\mathcal{O}_X$ -Algebra $r_*(\mathcal{O}_R)$, induces an equivalence between the category of finite flat coverings of S and the category of coherent and locally free $\mathcal{O}_X$ -Algebras. Let U be an open neighbourhood of Y , and let $r: R \rightarrow U$ be a finite flat covering of U . Let $\widehat{R}$ be the finite flat covering of $\widehat{X}$ deduced from it by change of base. One has $\widehat{r}_*(\mathcal{O}_{\widehat{R}}) \simeq r_*(\mathcal{O}_R)$.

Suppose then $\text{Lef}(X, Y)$. This implies that, for every U , the inverse image functor:

$$\mathbf{L}(U) \longrightarrow \mathbf{L}(\widehat{X})$$

is *fully faithful*. Indeed, let E and F be two locally free coherent $\mathcal{O}_U$ -Modules; $\underline{\text{Hom}}(E, F)$ is also coherent and locally free; $\widehat{\underline{\text{Hom}}}(E, F)$ is also coherent and locally free. By hypothesis, the natural map

$$\Gamma_U(\underline{\text{Hom}}(E, F)) \longrightarrow \Gamma_{\widehat{X}}(\widehat{\underline{\text{Hom}}}(E, F))$$

is an isomorphism, whence the conclusion, because $\underline{\text{Hom}}$ commutes with $\widehat{}$ since everything is locally free. Now $\widehat{}$ commutes with the tensor product, one deduces that the functor which to every coherent and locally free $\mathcal{O}_U$ -Algebra $\mathcal{A}$ associates the $\mathcal{O}_{\widehat{X}}$ -Algebra $\widehat{\mathcal{A}}$, is fully faithful. Better still, if E is a locally free coherent $\mathcal{O}_U$ -Module, there is a one-to-one correspondence between the structures of commutative $\mathcal{O}_{\widehat{X}}$ -Algebra on $\widehat{E}$.

PROPOSITION 2.3. *Let X be a locally noetherian prescheme and let Y be a closed subset of X . Let $\widehat{X}$ be the formal completion of X along Y . For every open U of X , $U \supset Y$, denote by L_U (resp. P_U , resp. E_U) the functor which to every locally free coherent $\mathcal{O}_U$ -Module (resp. to every finite flat covering of U , resp. to every étale covering of U) associates its inverse image under $\widehat{X} \rightarrow X$.*

- (i) *If one has $\text{Lef}(X, Y)$, then for every open neighbourhood U of Y , the functors L_U , P_U and E_U are fully faithful.*
- (ii) *If one has $\text{Leff}(X, Y)$, then for every locally free coherent $\mathcal{O}_{\widehat{X}}$ -Module $\mathcal{E}$ (resp....), there exist an open U and a locally free coherent $\mathcal{O}_U$ -Module E (resp....), such that $L_U(E) \simeq \mathcal{E}$ (resp....).*

²N.D.E. condition (ii) is superfluous, see the note on page 1.

(i) Has been seen.

(ii) Follows from (i) and the hypothesis, at least for L_U and P_U . Moreover if R is an étale covering of $\widehat{X}$, there exist an open neighbourhood U of Y in X and a finite flat covering R' of U such that $\widehat{R'} \simeq R$. One deduces from this a covering R'' of Y which is étale by 1.1, hence R' is étale in a neighbourhood U' of Y . qed

COROLLARY 2.4. *If one has $\text{Lef}(X, Y)$, for a finite flat covering R of an open neighbourhood U of Y to be connected, it is necessary and sufficient that $R \times_U \widehat{X}$ be so. In particular, for Y to be connected, it is necessary and sufficient that the open neighbourhood U of Y be so, or again that X be so.*

Indeed, for a locally ringed space $(X, \mathcal{O}_X)$ to be connected, it is necessary and sufficient that $\Gamma(X, \mathcal{O}_X)$ not be a direct product of two nonzero rings. Now one has

$$\Gamma(U, r_*(\mathcal{O}_R)) \simeq \Gamma(\widehat{X}, \widehat{r}_*(\mathcal{O}_{\widehat{R}}))$$

by $\text{Lef}(X, Y)$.

COROLLARY 2.5. *If one has $\text{Lef}(X, Y)$, then for every U , the functor*

$$\mathbf{Et}(U) \longrightarrow \mathbf{Et}(Y)$$

is fully faithful. If one has $\text{Lef}(X, Y)$, then for every étale covering R of Y , there exist an open neighbourhood U of Y and a covering R' of U such that $R' \times_U Y \simeq R$.

COROLLARY 2.6³. *If one has $\text{Lef}(X, Y)$ and if Y is connected, every open neighbourhood U of Y is connected and the natural homomorphism $\pi_1(Y) \rightarrow \pi_1(U)$ is surjective. If moreover one has $\text{Lef}(X, Y)$, the natural homomorphism*

$$\pi_1(Y) \longrightarrow \varinjlim_U \pi_1(U)$$

is an isomorphism. (N.B. One assumes chosen a “base-point” in Y , which one also takes as base-point in X , for the definition of the fundamental groups.)

All this follows trivially from prop. 1.1 and from prop. 2.3.

3. Comparison of $\pi_1(X)$ and of $\pi_1(U)$

DEFINITION 3.1. Let X be a prescheme and Z a closed subset of X . Set $U = X - Z$. One says that the pair (X, Z) is pure if, for every open V of X , the functor

$$\begin{aligned} \mathbf{Et}(V) &\longrightarrow \mathbf{Et}(V \cap U) \\ V' &\longmapsto V' \times_V (V \cap U) \end{aligned}$$

³**N.D.E.** together with 3.3 and the criteria ?? and ??, one obtains the following relative Lefschetz theorem. Let $f : X \rightarrow S$ be a projective and flat morphism of connected noetherian schemes and let D be an effective relative Cartier divisor in X which is relatively ample. If, for every $s \in S$, the depth of X_s at each closed point is ≥ 2 , then D is connected and, for every open U of X containing D , the arrow $i_U : \pi_1(D) \rightarrow \pi_1(U)$ is surjective. If moreover, the depth of X_s along each closed point of D_s is ≥ 3 and if the local rings of X at its closed points are pure, for example complete intersections (cf. X 3.4), then i_X is an isomorphism. Cf. Bost J.-B., “Lefschetz theorem for Arithmetic Surfaces”, *Ann. Sci. Éc. Norm. Sup. (4)* **32** (1999), p. 241–312, theorems 1.1 and 2.1. In the case where X is simply a smooth projective geometrically connected surface over a field, one always has connectedness of D and surjectivity of $\pi_1(D) \rightarrow \pi_1(U)$ (where U is an open containing D) for D merely nef of square > 0 (cf. *loc. cit.*, theorem 2.3 and also theorem 2.4 for surfaces that are merely normal and complete). In the case of an arithmetic surface (normal and quasi-projective) X over a ring of integers $\mathcal{O}_K$, Bost, improving results of Ihara (Ihara Y., “Horizontal divisors on arithmetic surfaces associated with Belyi uniformizations”, in *The Grothendieck theory of dessins d’enfants (Luminy, 1993)*, London Math. Soc. Lect. Note Series, vol. 200, Cambridge Univ. Press, Cambridge, 1994, 245–254 or *loc. cit.*, corollary 7.2), has shown that if a point $P \in X(\mathcal{O}_K)$, which plays the role of the divisor D in the geometric situation, satisfies certain positivity conditions, then the arrow $\pi_1(X) \rightarrow \pi_1(\text{Spec } \mathcal{O}_K)$ deduced from the projection was invertible, with inverse the arrow $\pi_1(\text{Spec } \mathcal{O}_K) \rightarrow \pi_1(X)$ deduced from P (*loc. cit.*, theorem 1.2).

is an equivalence of categories⁴.

DEFINITION 3.2. Let A be a noetherian local ring. Set $X = \text{Spec } A$. Let $\mathfrak{r}(A)$ be the radical of A and let $x = \mathfrak{r}(A)$ be the closed point of X . One says that A is pure if the pair $(X, \{x\})$ is so.

We leave to the reader the task of not proving the following proposition:

PROPOSITION 3.3. *Let X be a locally noetherian prescheme and let Z be a closed subset of X . For the pair (X, Z) to be pure it is necessary and sufficient that, for every $z \in Z$, the ring $\mathcal{O}_{X,z}$ be pure⁵.*

This being said the following theorem is the essential result of this number:

THEOREM 3.4 (PURITY THEOREM⁶). (i) *A regular noetherian local ring of dimension ≥ 2 is pure (purity theorem of Zariski–Nagata).*

(ii) *A noetherian local ring of dimension ≥ 3 which is a complete intersection is pure.*

Let us recall that one says that a local ring is a *complete intersection* if there exists a regular noetherian local ring B and a B -regular sequence $(t_1, \dots, t_k)$ of elements of the radical $\mathfrak{r}(B)$ of B such that

$$A \simeq B/(t_1, \dots, t_k).$$

In this connection let us remark that it would be less ambiguous to say that A is an *absolute complete intersection*, as opposed to the situation, which we have already encountered, where X is a locally noetherian prescheme (which need not be regular) and where Y is a closed subset of X , of which one says that it is “set-theoretically locally a complete intersection in X ”.

Let us first prove some lemmas.

LEMMA 3.5. *Let X be a locally noetherian prescheme and let U be an open subset of X . Set $Z = X - U$. Let $i: U \rightarrow X$ be the canonical immersion of U in X . The following conditions are equivalent:*

- (i) *For every open V of X , if one sets $V' = V \cap U$, the functor $F \mapsto F|_{V'}$ from the category of locally free coherent $\mathcal{O}_V$ -Modules to the category of locally free coherent $\mathcal{O}_{V'}$ -Modules is fully faithful;*
- (ii) *the natural homomorphism $\mathcal{O}_X \rightarrow i_*(\mathcal{O}_U)$ is an isomorphism;*
- (iii) *for every $z \in Z$, one has $\text{prof } \mathcal{O}_{X,z} \geq 2$.*

One has already seen (?? ??) the equivalence of (ii) and (iii). Let us show that (ii) implies (i). Let F and G be two locally free coherent $\mathcal{O}_V$ -Modules, $\underline{\text{Hom}}(F, G)$ is so as well, hence $\underline{\text{Hom}}(F, G) \rightarrow i_*(\underline{\text{Hom}}(F|_{V'}, G|_{V'}))$ is an isomorphism hence $\text{Hom}(F, G) \simeq \text{Hom}(F|_{V'}, G|_{V'})$. Conversely one takes $F = G = \mathcal{O}_X$ and one applies (i) to every open V of X .

Here is a useful “descent lemma”:

LEMMA 3.6. *Let X be a locally noetherian prescheme and let Z be a closed subset of X . Set $U = X - Z$. Suppose that the homomorphism $\mathcal{O}_X \rightarrow i_*(\mathcal{O}_U)$ is an isomorphism. Let $f: X_1 \rightarrow X$ be a faithfully flat and quasi-compact morphism. Set $Z_1 = f^{-1}(Z)$. If the pair (X_1, Z_1) is pure, so is (X, Z) .*

Let us remark that the hypothesis $\mathcal{O}_X \simeq i_*(\mathcal{O}_U)$ is preserved by flat extension of the base, because i is a quasi-compact morphism and, in this case, direct image commutes with inverse image. Now this hypothesis implies that the functor

$$\mathbf{Et}(V) \longrightarrow \mathbf{Et}(U \cap V)$$

⁴For a notion more satisfactory in certain respects, cf. the comment ?? ?? d).

⁵Compare the non-commutative case of ?? ??, whose proof is essentially the same as that of 3.3.

⁶**N.D.E.** for the history of the methods employed, see the letter of 1 October 1961 from Grothendieck to Serre, *Correspondance Grothendieck-Serre*, edited by Pierre Colmez and Jean-Pierre Serre, Documents Mathématiques, vol. 2, Société Mathématique de France, Paris, 2001.

defined by

$$V' \longmapsto V' \times_V (V \cap U)$$

is fully faithful, as is shown by the interpretation of an étale covering in terms of a locally free coherent Algebra. It remains to prove effectivity. One may for example introduce the square X_2 and the cube X_3 of X_1 over X and one remarks that a faithfully flat and quasi-compact morphism is a morphism of *universal effective descent* for the fibered category of étale coverings, over the category of preschemes. The conclusion is formal from there⁷.

REMARK 3.7. One has proved along the way that if $\mathcal{O}_X \rightarrow i_*(\mathcal{O}_U)$ is an isomorphism, X is connected if and only if U is so, and then $\pi_1(U) \rightarrow \pi_1(X)$ is surjective.

COROLLARY 3.8. *Let A be a noetherian local ring. Suppose that $\text{prof } A \geq 2$. Then if $\widehat{A}$ is pure, A is pure.*

Follows from lemma 3.5 and lemma 3.6.

The following lemma is the essential point of the proof of the purity theorem:

LEMMA 3.9. *Let A be a noetherian local ring and let $t \in \mathfrak{r}(A)$ be an A -regular element. Suppose that A is complete for the t -adic topology and moreover a quotient of a regular local ring (for example A complete). Set $B = A/tA$.*

- (i) *If for every prime ideal $\mathfrak{p}$ of A such that $\dim A/\mathfrak{p} = 1$, one has $\text{prof } A_{\mathfrak{p}} \geq 2$, then B pure implies A pure.*
- (ii) *If for every prime ideal $\mathfrak{p}$ of A such that $\dim A/\mathfrak{p} = 1$, one has $\text{prof } A_{\mathfrak{p}} \geq 2$, if $A_{\mathfrak{p}}$ is pure when $t \notin \mathfrak{p}$, and if⁸ $\text{prof } A_{\mathfrak{p}} \geq 3$ when $t \in \mathfrak{p}$, then A pure implies B pure.*

Let $X' = \text{Spec}(A)$ and let $Y' = V(t)$, which one identifies with the spectrum of B . Let $x = \mathfrak{r}(A)$, and set $X = X' - \{x\}$ and $Y = Y' - \{x\} = X \cap Y'$. Let us denote by $\widehat{X'}$ the formal spectrum of A for the t -adic topology, which is identified with the formal completion of X' along Y' .

Since A is complete for the t -adic topology, one remarks that $\mathbf{Et}(X') \rightarrow \mathbf{Et}(\widehat{X'})$ is an equivalence of categories. Likewise $\mathbf{Et}(\widehat{X'}) \rightarrow \mathbf{Et}(Y')$ by prop. 1.1, hence $\mathbf{Et}(X') \rightarrow \mathbf{Et}(Y')$ is an equivalence of categories.

Let us show (i). Consider the diagrams

$$\begin{array}{ccc} X' & \longleftarrow & X \\ \uparrow & & \uparrow \\ Y' & \longleftarrow & Y \end{array} \qquad \begin{array}{ccc} \mathbf{Et}(X') & \xrightarrow{a} & \mathbf{Et}(X) \\ \downarrow c & & \downarrow b \\ \mathbf{Et}(Y') & \xrightarrow{d} & \mathbf{Et}(Y) \end{array}$$

One has just seen that c is an equivalence, d is so as well by the hypothesis that B is pure and finally b is fully faithful as one has seen in example 2.1, cf. 2.3 (i).

Let us show (ii). This time one assumes that A is pure hence a is an equivalence; likewise c . Let us see that b is an equivalence. By example 2.1 one knows that one has $\text{Leff}(X, Y)$, hence b is already fully faithful, let us prove that it is essentially surjective. One uses 2.3 (ii) noting that, if U is an open neighbourhood of Y in X , the complement of U in X is a union of finitely many closed points; the pair $(X, X - U)$ is therefore pure by prop. 3.3, because at such a point $\mathfrak{p}$, $\mathcal{O}_{X, \mathfrak{p}} = A_{\mathfrak{p}}$ is pure by hypothesis. Whence the conclusion.

PROOF OF THE PURITY THEOREM. Let us first show (i) by induction on the dimension. Let A be a noetherian local ring of *dimension* 2. Set $X' = \text{Spec}(A)$, $x = \mathfrak{r}(A)$, $X = X' - \{x\}$. One has $\text{prof } A = 2$. One can therefore apply lemma 3.5 to the pair $(X', \{x\})$ and hence $\mathbf{Et}(X') \rightarrow \mathbf{Et}(X)$ is fully faithful. Let now $r: R \rightarrow X$ be an étale covering defined by a

⁷Cf. J. Giraud, *Méthode de la descente*, Mémoire no. 2 of the Bulletin de la Société Mathématique de France (1964).

⁸N.D.E. this last condition can be improved, cf. the editor's note (1) on page 2.

locally free coherent and étale $\mathcal{O}_X$ -Algebra $\mathcal{A} = r_*(\mathcal{O}_R)$. Let us denote by $i: X \rightarrow X'$ the canonical immersion of X in X' . I say that $i_*(\mathcal{A}) = \mathcal{B}$ is a *coherent* $\mathcal{O}_{X'}$ -Algebra. Indeed, it suffices to apply the “finiteness theorem” ?? ???. I say that this algebra is of depth ≥ 2 at x . Indeed, it is the direct image of an $\mathcal{O}_X$ -Module, with $X = X' - \{x\}$. Since A is a *regular* ring of dimension 2 one has: $\text{dp } \mathcal{B} + \text{prof } \mathcal{B} = \dim A = 2$, where $\text{dp } \mathcal{B}$ denotes the projective dimension of $\mathcal{B}$. Hence $\text{dp } \mathcal{B} = 0$, hence $\mathcal{B}$ is projective, hence free. It follows that $\mathcal{B}$ defines a finite flat covering of $X' = \text{Spec}(A)$. The set of points of X' where this covering is not étale is a closed subset of X' whose equation is a principal ideal: the discriminant ideal of $\mathcal{B}/A$. Now, by construction, this closed set is contained in $x = \mathfrak{r}(A)$ hence is empty because $\dim A = 2$.

Let A be a regular noetherian local ring, $\dim A = n \geq 3$. Suppose (i) proved for rings of dimension $< n$. To prove that A is pure, one may assume A complete by 3.8. Let $t \in \mathfrak{r}(A)$ whose image in $\mathfrak{r}(A)/\mathfrak{r}(A)^2$ is nonzero. Then $B = A/tA$ is a *regular* noetherian local ring of dimension $n - 1$ hence is pure, because $n - 1 \geq 2$. One concludes by lemma 3.9 (i), which is applicable because A is complete.

Let us show (ii). Let A be a noetherian local ring of dimension ≥ 3 . Suppose there exists a *regular* noetherian local ring B and a B -sequence $(t_1, \dots, t_k)$ such that $A \simeq B/(t_1, \dots, t_k)$. Let us prove that A is pure, by induction on k . If $k = 0$, one knows it by (i). *Suppose* $k \geq 1$ *and the result acquired for* $k' < k$. By corollary 3.9 one may assume that A (hence also B) is complete. Set $C = B/(t_1, \dots, t_{k-1})$, hence $A \simeq C/t_k C$ and t_k is C -regular. By the induction hypothesis one knows that C is pure, it suffices to prove that lemma 3.9 (ii) is applicable. Notation: A and B of the lemma become C and A . One has $\dim C \geq 4$, hence for every prime ideal $\mathfrak{p}$ of C such that $\dim C/\mathfrak{p} = 1$, one has $\text{prof } C_{\mathfrak{p}} \geq 3$. Moreover, $C_{\mathfrak{p}}$ is a complete intersection with $k' \leq k - 1$, hence is pure by the induction hypothesis. $\square$

THEOREM 3.10. *Let X be a locally noetherian prescheme and let Y be a closed subset of X . Suppose that one has $\text{Leff}(X, Y)$ (cf. Examples 2.1 and 2.2). Suppose moreover that, for every open neighbourhood U of Y and every $x \in X - U$, the local ring $\mathcal{O}_{X,x}$ is regular of dimension ≥ 2 or else a complete intersection of dimension ≥ 3 . Then*

$$\pi_0(Y) \longrightarrow \pi_0(X)$$

is a bijection, and if X is connected

$$\pi_1(Y) \longrightarrow \pi_1(X)$$

is an isomorphism.

There is nothing left to prove. One remarks that, in the two examples cited 2.1 and 2.2, the complement of U is a union of finitely many closed points, whence it follows that the hypothesis on the dimension of $\mathcal{O}_{X,x}$ is not a joke.